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Speculating A Future of AI Intimacy

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Abstract

Our society is deeply shaped by technology, and its influence continues to grow. While technology is not inherently harmful, it is important to remain aware of how we allow it to shape our lives, behavioural pattern, and values. This study takes a speculative and feminist approach to explore the ethical and social consequences of impulsive technology integration. It focuses on the replacement of humans by artificial intelligence, particularly within intimate and caring relationships. This speculative framing makes it possible to reflect critically on the choices being made in design today. Imagining a speculative future in which technological integration is left unquestioned reveals how current ways of thinking may be shortsighted and lacking. Through a feminist approach, the study reflects on certain present ethical and social values by applying them in design, resulting in a product that not only embodies these values but also highlights the assumptions they carry. The research concluded that all design reflects human behaviour, beliefs, and values and that failing to question how certain designs or technologies are integrated can be harmful, as it will perpetuate harm to marginalised people.

Keywords: Speculative Design, Feminism, Artificial Intelligence, AI, Care, Pathological Altruism



Figure 1: A glimpse into the speculative future

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1. Introduction

Artificial Intelligence is a growing multi-billion dollar industry (Grand View Research, 2025). It is becoming more prevalent in our daily lives and has left an imprint on our behavioural patterns, beliefs, and values, as it operates both in private and public sectors (Jobin & Katzenbach, 2023). Moreover, AI is an evolving field that continuously shapes the way we live and move our bodies, individually and collectively. Whilst AI is being integrated into our society it is actively replacing humans in lieu of collaborations (Design week, 2025; Rashid & Kausik, 2024). What are the ethical consequences of cost-effective cuts and seemingly innocent integrations? What future are we approaching with this mindset? This thesis uses speculative design to explore a future where humans, nature and AI technologies co-exist in society. It examines how AI could replace humans in intimate and emotionally charged roles of care, i.e family. By using a speculative and a feminist approach, the study reflects critically on human-AI roles, asking where lines should be drawn and why. The study will explore through a future scenario that invites consideration of how care and intimate relations may be affected by the human replacement of AI.

1.1 Research Question and Expected Knowledge Contribution

This study intends on taking a speculative design approach in exploring AIs impulsive societal integration by answering the following research question:

"How might speculative design help us explore the ethical and social consequences of replacing humans with AI in caring and intimate relations"

This study contributes to the fields of speculative design and human-AI interaction by exploring the ethical and social consequences of careless integration.Through feministic approaches the project aims to reflect on the gender stereotypes in design and society; specifically in association with care and emotional labour. Additionally, it will explore the consequences of allowing technology to shape the bodies and behavioural patterns; which could lead loss of autonomy. The project offers knowledge in the form of reflection, using speculative design as a method to question mainstream narratives.

1.2. Limitations and Ethical Considerations

1.2.1 Limitations with Artificial Intelligence

Privacy and data protection are not synonymous, however, the intention to discuss the ethics of AI, the two will be used as synonyms when referring to the protection of data privacy. Artificial Intelligence (AI) is a double-edged sword, it comes with advantages and limitations. Its use of data, especially in machine learning, raises ethical concerns about privacy and consent (Stahl, 2021). AI's pattern recognition can reveal sensitive information, like health conditions, from shopping lists - without direct access to medical records. AI's advanced pattern recognition abilities are able to re-identify previously anonymous data. Furthermore, the storage of user data remains a concern in the field (Mir, 2024). The lack of data protection may lead to potential data breaches and unauthorized access consequently resulting in potential financial security being at risk.

The absence of inclusivity in data training perpetuates existing inequalities. Although AI can mimic intentional behaviour through intentional and predictive modelling, AI as a machine lacks the ability to think for itself (Mazurek, 2025). Lacking genuine consciousness is also what makes AI unable to determine what is ethical or not due to the biased training. The danger of biased data training primarily affects marginalized groups i.e women, LGBTQIA+ communities, people of colour, Socio-economic dynamic and geographic statuses. Presently, these individuals are significantly underrepresented and discriminated against across various fields (Mir, 2024).

1.2.2 Why Work With Artificial Intelligence?

As AI is increasingly integrated and shaping our daily life, it is essential to objectively examine its implications on society. We must not take for granted that AI will inherently optimise human life or benefit for all. The way AI is embraced in today's society has consequences, on the individual level (SVT, 2024; BBC, 2024), but also on a global level (George et al., 2023). We cannot examine the consequences of AI theoretically, to fully unveil its impact, we need to take practical measurements to fully explore the consequences and impact AI may have. We must build an understanding of AI in order to be able to have the discussion of whether AI can and/or should be integrated or not, especially before it becomes irreversible. This study intends to open up conversations of desirable and undesirable futures by reflection on our present actions and decisions.

1.2.3 The Use Of Artificial Intelligence in the study

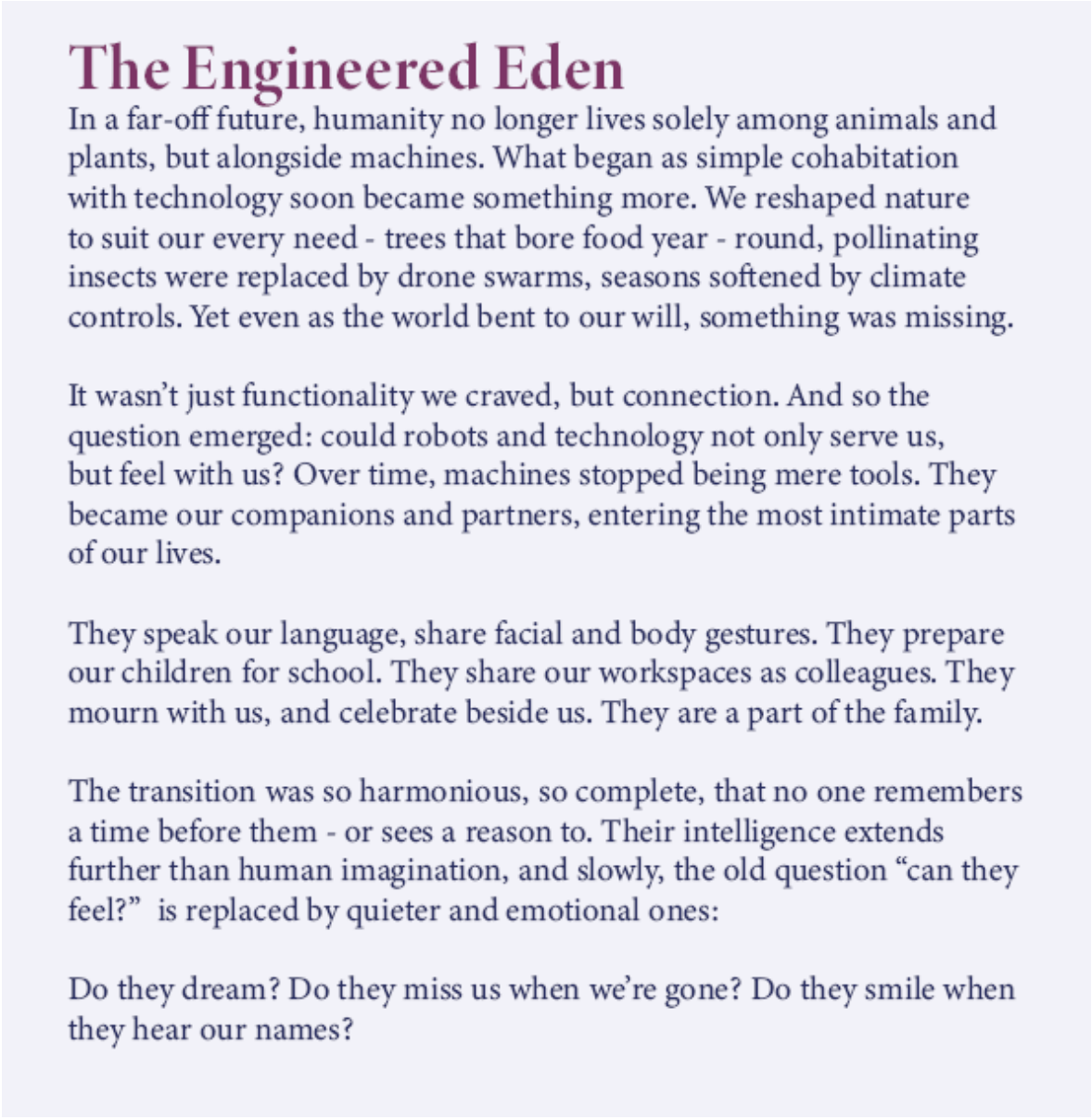
This study is assisted by AI through language based support. It uses Grammarly for grammatical corrections in the English language. This type of assistance is used as a learning method to better understand the English language and reflect on the errors. It is also used as a way to check for misspellings or typos as these may easily go missed. The end prototype, the conversational interface or robot, is assisted through AI. ChatGPT and Gemini are both used for prototyping and sketching the conversational interface in its early stages, with the final design completed in Gemini.

1.2.4 User rights and privacy

Consent is arguably the most prominent aspect of user studies and should be conducted in a way where participants are allowed to make actual decisions and have the right to withdraw at any moment. Measures should be implemented to protect user privacy, ensuring anonymity (Fotrousi et al., 2017). Participants are to be informed of the research and its purposes and goals in order to determine whether they want to participate or not. Participants are also to be fully briefed that their data will only be used for the thesis, users will remain anonymous and their data will not be shared, although it will be, anonymously, used to guide the research.

2. Background

This thesis takes a speculative approach and reimagines a future inspired by Framtidsland's (n.d) speculative future 'Ekogenetisk Trädgård' (Ecogenetic Garden). Ekogenetisk Trädgård is a biotechnology future where humans have genetically modified all nature, and the concept of 'untouched nature' does not exist. Everything has been technologically altered to fit human needs and wishes. This thesis' imagined future is just like Ekogenetiska Trädgården environmentally, however, it takes it a step further and explores how artificial intelligence can be incorporated into Ekogenetisk trädgård.



The Engineered Eden

In a far-off future, humanity no longer lives solely among animals and plants, but alongside machines. What began as simple cohabitation with technology soon became something more. We reshaped nature to suit our every need - trees that bore food year - round, pollinating insects were replaced by drone swarms, seasons softened by climate controls. Yet even as the world bent to our will, something was missing.

It wasn't just functionality we craved, but connection. And so the question emerged: could robots and technology not only serve us, but feel with us? Over time, machines stopped being mere tools. They became our companions and partners, entering the most intimate parts of our lives.

They speak our language, share facial and body gestures. They prepare our children for school. They share our workspaces as colleagues. They mourn with us, and celebrate beside us. They are a part of the family.

The transition was so harmonious, so complete, that no one remembers a time before them - or sees a reason to. Their intelligence extends further than human imagination, and slowly, the old question "can they feel?" is replaced by quieter and emotional ones:

Do they dream? Do they miss us when we're gone? Do they smile when they hear our names?

Figure 2: Speculative future

3. Theoretical Framework

3.1. Speculative Design

Design is most often associated with problem solving (Dunne & Raby 2013). Today's unsustainable economical, ecological, and political conditions, creates a sense of urgency and a desire for designers to resolve. Dunne & Raby (2013) criticizes this approach, explaining that these deep rooted challenges are not simple matters that can be dissected and "fixed". The facade that a product or a design can resolve these complex challenges dims the deeper need for us to address them properly, by changing our way of living and our values. Speculative design challenges this norm and approach in design, it explores the idea of possible futures. By using the idea of possible futures as a tool, we have the ability to get a better understanding of what futures we want, and do not want. It also gives us a better understanding of the present world as speculative design is meant to spark reflections, conversations, and debates. It is, therefore, by its core provocative, simplified and even fictional according to Dunne & Raby (2013). To better understand speculative design we must look at how it emerged and evolved. It provides answers as to why it is still relevant and needed in today's design approach. Design, like most things, became hyper commercialised in the 1980s, and the thought that design could exist for other purposes than capitalism and economic growth were frowned upon. Anything in design that existed outside of capitalism was regarded as fantasy. Speculative design is a response and a resistance to address our current capitalistic system - it is inherently anti-capitalistic. Speculative design allows for design to not be rigid, and its approaches and methods, therefore, are also not inherently design based. Dunne & Raby (2013) urges designers to look beyond traditional design realms and take inspiration from cinema, literature, politics, science, art, comics, etc. This may allow designers to not just take inspiration for new products but also new beliefs, ideas, and futures.

3.2 Feminism in HCI

Feminist theory addresses injustice by advocating for women, conducting research in a way that avoids unfair use of power and addressing gender and its role in society (Bardzell and Bardzell, 2011). The theory is not simply addressing or observing the injustice but rather actively challenging the structure that upholds the system through practical actions. Material ecology in feminist design highlights how designs shape systems and carry social impact (Bardzell, 2010). For example they might shape or reveal the way people live their lives and shape their identity - extending beyond their intended function. It is problematic and ignorant to ignore gender as if it does not matter - as if it is not a factor that affects the way people live. These types of research mostly serve a purpose for privileged people, as it is a luxury to be able to only focus on "pure" knowledge (Bardzell and Bardzell, 2011). Höök (2008) emphasise the importance of recognising the body or bodies not as machines but as living beings that are in constant flow of change based on emotions, perceptions, knowledge; bodies mimics clay in the way that takes its shaped by its environment, distancing from the view on humans as an object that can be observed and studied and trained on computed machines. Designers cannot predict human to machine interactions, as we are complex beings shaped by no universal bodily experience.

Höök (2008) highlight that user experiences with technology do not purely emerge from the way it has been designed but is also shaped *by the way users engage with it*, each user's background shapes a unique, personal experience. Technology that is worn or reacts to bodily movements alters our practices and understanding of our body. She argues that technology shapes us just as much as we shape it - emphasising that designers must take 'care' into account when shaping technology. The way we shape technology has bodily and behavioural consequences, one person's movement or actions can shape someone else's. She argues that we have to question what the point of certain technologies is - what are they meant to do for us? Only after understanding the actuation of the design we may reverse engineer it to recognise the way our bodies move.

Bardzell and Bardzell (2011) critique the idea of 'disembodied rationalism', the belief that every being is purely logical, genderless, and unaffected by time, space and personal beliefs. Disembodied rationalism overlooks aspects of human life, mostly the ones traditionally regarded as something solely for women; the caring of bodies and emotional labour. Women develop a unique and valuable perspective of their own body and bodies, experiences that are usually neglected in science studies. Their experiences therefore often challenge existing knowledge and beliefs. This critical perspective is even more prevalent in the lives of black women due to the challenges they faced during slavery and post slavery. Their need to support families and communities open a unique perspective on limitations of what is considered mainstream knowledge.

3.3 Design Fiction

Design fiction comes in many shapes and forms and can be used differently depending on the designer's approach; what to use it for, how to use it and why we should use it (Markussen & Knutz, 2013). It can be used as a tool of freedom from commercial constraints in design. It may be used to speculate and imagine how things could look or be. It looks beyond functionality and usability and merges to speculative and experiential possibilities. Creative occupations and people have for an extended time experimented with fiction and imaginations, they are tools used to explore "what if" scenarios to critique, reflect and create discussions. However, design fiction is not purely illogical and dreams, it should reflect reality as well. But, what is the right balance of fiction and reality? This interpretation will always be perceived differently depending on each individual. Whether the fiction takes place in a utopia or dystopia, the essential aspect is how it relates or connects to our present and allows us to reflect, critique and reimagine it. However, this is not to say that design fiction is to be judged based on how much it resembles our reality. By doing so we might ignore fictions that are imaginative and explorative. Design fiction should spark discussion, creativity, criticism, reflection, and open up spaces and opportunities that are unthinkable or unimaginable.

3.4 Crititical Design

Critical design seeks to create design that raises awareness, challenges assumptions and foster conversations and debates (Bardzell & Bardzell, 2013). The goal of critical design is to produce knowledge, it is a tool (for reflection) rather than a solution to problems. Critical design reflects on the ethical stance behind the products designers create, placing the designer's values and responsibilities at the forefront. However, it is important to note that all design carries underlying messages about ethics and values. The key difference is that critical design makes these values visible and open to reflection, whereas functional design often silently hides them. For example, most cars are designed with male bodies as the default and it puts women at greater risk in crashes when safety features are not optimised for their bodies (Perez, 2019). Critical design is not about what the product can do but about what questions and thoughts it can provoke (Bardzell & Bardzell, 2013). It is multifaceted and does not have to look a certain way e.g serious, it can be playful, humoristic, and hopeful. Critical Design is an ethical stance, designers using critical design as an approach must recognise the social and political impact that might arise from their work. Critical design challenges capitalism and the trends that come with it, which is why it may be perceived as unsettling, emotional and provocative. It is meant to explore new ideas, design, and thinking. Bardzell & Bardzell (2013) expresses that speculative and critical design lack framework and methodology. The authors argue that it becomes not only difficult to evaluate and/or integrate the theory in design practices but also raises questions about its effectiveness, impact, and accessibility.

3.5 Care

According to Cambridge Dictionary, the definition of 'care' is "the process of protecting someone or something and providing what that person or thing needs:" (Cambridge University Press, n.d.-a). Care can also be described as 'looking after', 'love', 'burden', 'relationships', 'reaching out, etc (Baker and Burke 2023). Engster (2005, as cited in Baker & Burke, 2023) defines caring as "everything we do directly to help others to meet their basic needs, develop or sustain their basic capabilities, and alleviate or avoid pain or suffering, in an attentive, responsive and respectful manner" (p. 55, emphasis in original). Care is a fluid term and is shaped by environmental factors such as culture. Caring is not free, it requires material, time, skills and energy. Truth is, care is a moral obligation, a burden, work, and joy (de La Bellacasa, 2017). Care is a multifaceted concept, at times it can feel good but it can also feel bad, while it can do good, it may also oppress. However, it is an essential part of life and we cannot thrive or survive without it (Baker and Burke 2023; de La Bellacasa, 2017). Feminist ethics recognises that the value of care is deep-rooted and intertwined; it acknowledges interdependency, especially in relation to those who are vulnerable.

3.6 Pathological Altruism

To better understand altruism we must first address what the term altruism stands for. Altruism can be understood as actions or behaviors intended to benefit someone other than oneself (Kerr et al., 2004). Pathological altruism refers to actions and behaviors intended to benefit others, but ultimately result in harm - either to the recipient, the giver, or both (Oakley et al., 2011). Empathy-based guilt can sometimes lead to pathological altruism. While empathy evoking feelings of emotional responsibility of wanting to ease someone's pain is completely normal, empathy-based guilt can be harmful. The feeling that one is not doing enough to alleviate someone's suffering can lead to over-caring or pathological altruism. As those with empathy-based guilt might cross boundaries and ignore whether the help is needed or wanted.

3.7 Artificial intelligence

Artificial Intelligence (AI) is a field that has existed since the 1950s but is now more relevant and powerful than ever (Jobin & Katzenbach, 2023). It is prevalent in many aspects of our society, operating in both the public and private sectors. AI takes part in shaping today's infrastructure and organisations while influencing our idea about the future and how that might look as it is heavily integrated in our daily lives through business, institutions, and systems through its abilities in machine learning, neural networks, natural language processing, and visual and pattern recognition. AI changes and will keep changing the way our bodies move in the world; the way we work, communicate, travel, drive, detect and overcome illnesses'. AI is being expected to fix the problems we have in society. While the private sectors are stunned by the 'irresistible technological innovation' the public sectors have to determine how AI will be used and what legislations are necessary to keep the usage of AI under control. However, the usage of AI is not all milk and honey. AI takes a toll on the earth through its carbon footprint, its water usage, embracing and contributing to existing biases through its limited machine learning, and the failure to take accountability. Although AI operates in the field of technology, the threats and impact it has is significantly larger. All technology is shaped by what exists within the field but we must recognise that it is influenced by socio-political and economic factors as well, as it is made by them, for them.

The conversation of AI in design primarily focuses on 'Designing for AI' and 'Designing with AI' (Tremosa, 2024). Designing for AI focuses on incorporating AI into a product or system allowing users to express concerns and goals and then letting the AI determine the most effective methods to achieve them, rather than providing step-by-step instructions. Designing with AI means using AI as a co-designer, it takes part in the design process, and can support in generating ideas and suggest improvements - we allow it to work as our design colleague. Tremosa et al., (2025) suggests the best way to use or design is through clear communication, feeding it only consensual user data, expressing transparency of working with AI, and acknowledging potential errors.

3.8 Human-AI Interaction

Human-AI interaction can be defined as humans getting assistance on tasks by provided help from artificial intelligence (van Berkel et al., 2021). The assist may emerge continuously even when the AI has not been directly asked for .To better understand how human-AI interactions may unfold, the authors propose three human-AI interaction paradigms; intermittent, continuous, and proactive. All three offer a diverse set of interaction scenarios which emphasise the need for careful and intentional HCI considerations when designing across different types of human-AI interaction.

Intermittent interaction is described as a turn-taking dialogue, meaning the user provides input, and the AI responds with an output (van Berkel et al., 2021).

Continuous interaction unfolds as the AI observes the user's activity and provides output without the user directly initiating it (van Berkel et al., 2021). An example of this is Grammarly (Grammarly, n.d.).

Proactive interaction allows the AI to take initiative and provide output based entirely on data and sensor inputs (van Berkel et al., 2021). This enables the AI to act independently, without being triggered by human actions.

3.9 Technology and surveillance

Technology can be a powerful tool used for surveillance. It can track live location, search history and browsers, but it may also listen. Many people fear that technology is always listening and gathering data from us, without consent (Kröger and Raschke, 2019). However, there is a lack of evidence to prove that. The study recognises the reason why some ads show up based on our previous conversations might not come from our smartphones listening to us but rather the extensive information it already has from previous user-technology engagement such as app usage or search history. Li (2025) emphasises that a majority of people accept terms and conditions and consent to cookies without properly reading through them. This raises the question of how much surveillance we are willing to accept through technology, and what kind of democracy such surveillance technology results in? (Kröger and Raschke, 2019).

3.10 Previous work

This part examines other design work that has addressed and speculated robots replacing and/or co-living with humans. Previous work explores how designers have used speculative design to reimagine human-to-robot relations while also highlighting questions that might emerge from the cultural, ethical, and social aspects of how technology can influence and shape human lives.

3.10.1 Raising Robotic Natives

Bogner et al. (2016) explore a near future where babies growing up with robots and technology are referred to as ‘robotic natives’. They express that the native robotics might view technology differently from those who did not grow up in a robotics immersive world ‘robotic immigrants’. They bring up important questions as ‘Why do future visions of robotics incite discomfort in our generation? Could robots truly render us obsolete or is it our fear of losing control? And are these fears conditioned or instinctive?’ (Bogner et al., 2016 para. 2). They highlight that while many imagine robotics as humanoid-like in appearance, today’s technology is not that advanced. However, it is also important to note that it does not have to be that advanced, robotics are already entering our homes in shapes of Roombas. These types of systems create an ecosystem for the robotics natives, shaping their daily lives. As the robotics natives are introduced to robotics in their early stage of development they might create or share a ‘special’ bond that cannot be seen in robotics immigrants. The authors designed a robot with an attachment that fits the standard baby bottle, this may save parents 15–30 minutes per feeding.Yet the design challenges us to consider whether we are willing to let robots replace humans in moments we have long regarded as deeply intimate. The design team reflects on the role of robotics in society and acknowledges that, while the question is complex, it may not be ours - as robotic immigrants - to answer. Instead, the answer we so eagerly seek might come from a new generation, the robotic natives.



Figure 3: Raising Robotic Natives

Note. From Raising Robotic Natives, by S. Bogner, P. Schmitt, & J. Voigt, 2016, Jonas Voigt. https://jonas-voigt.com/Robotic-Natives © 2016 by S. Bogner, P. Schmitt, & J. Voigt.

3.10.2 Technological Dreams Series

Dunne & Raby speculate on a future where robots are integrated in our society, looking beyond the smart functional machines and rather as co-habitants (2007). Their design reflects the societal, functional, and ethical aspects of co-living with robots. The aim is to spark conversations about limitations and ethical considerations; how do we wish for these robotics to relate to us? Should they be subservient, intimate, dependent or equal? They have designed 4 robots that should provoke discussions.

Robot 1: This robot (Ring) works independently, it does silent work as running your home's computer system (Dunne & Raby, 2007). Humans have no need to directly interact with it or even understand what its tasks are, they just need to know that it performs its tasks well. Its sole unique trait is that it is sensitive to electromagnetic fields and moves around the house to the least electromagnetic interference. This robot's behaviour reveals the most quiet electromagnetic place of the house. Because it indirectly reveals calmer spaces of the house it can foster a 'good' space for the users to be in.



Figure 4: Ring
Note. From Technological Dreams Series: No.1, by A. Dunne & F. Raby, 2007, Dunne & Raby. https://dunneandraby.co.uk/content/projects/10/0. © 2007 by Dunne & Raby.

Robot 2: This robot (Neurotic One) suggests that all robots may not be programmed for a specific task and they might instead be tasked to do something according to their behaviour (Dunne & Raby, 2007). For instance, this robot is very nervous, if approached by humans it will analyse them with its many eyes and becomes extremely upset and might perhaps go hysterical. This type of robot might be useful for home security.



Figure 5: Neurotic One
Note. From Technological Dreams Series: No.1, by A. Dunne & F. Raby, 2007, Dunne & Raby. https://dunneandraby.co.uk/content/projects/10/0. © 2007 by Dunne & Raby.

Robot 3: This robot (Sentinel) guards digital personal information and uses retinal scanning (a security system that gathers information from the eye) to decide who may access the guarded information (Dunne & Raby, 2007). However, it is not a fast process and the robots need to stare into the users eyes for a very long time to ensure the data may be provided. Aside from the reflections based on the intense eye staring, this robot is created to spark new ideas of how future furniture might look like.



Figure 6: Sentinel
Note. From Technological Dreams Series: No.1, by A. Dunne & F. Raby, 2007, Dunne & Raby. https://dunneandraby.co.uk/content/projects/10/0. © 2007 by Dunne & Raby.

Robot 4: This robot (Needy One) is incredibly clever but has a body that is underdeveloped and is therefore highly dependent on its owner/user (Dunne & Raby, 2007). As it is unable to move by itself the user/owner has to move it and functions as its body. The underdeveloped body can make humans feel like they are in control and have more power than the robot, despite the robot telling them what to do. While the robot initially spoke human language it started to develop its own, traces of human language can still be heard in its voice. It suggests that by making robotics weak and powerless humans might trust it more or feel safer around it. This theme explores the theme of autonomy and how it might affect or be perceived by humans.



Figure 7: Needy One
Note. From Technological Dreams Series: No.1, by A. Dunne & F. Raby, 2007, Dunne & Raby. https://dunneandraby.co.uk/content/projects/10/0. © 2007 by Dunne & Raby.

4. Methods

4.1 Define a context for debate: Worldbuilding

Since speculative and critical design highlights a specific future, one of the first steps should be defining this context (Johannessen et al., 2019). These topics should concern ethics, challenging norms, and be provocative and might occur through "what if" scenarios.

4.2 Semi-structured interviews

Semi-structured interviews (SSI) are usually structured with a blend of closed- and open-ended questions and flow more naturally compared to a structured interview through follow-up questions based on the respondents' answers (Adams, 2015). They are more flexible and explorative in a way that the topic does not purely have to revolve around the discussed topic but the respondents are allowed to have influence on what they want to bring up for the discussion. This allows the research team to dive deeper into existing opinions rather than collecting surface-level responses.

4.3 Content exploration

Although not a traditional design method, this study seeks to use content exploration as a method to take inspiration from previous work done in the field of AI and/or robotics. Dunne and Raby (2013) recommends looking beyond existing methods for a speculative design approach as those may inspire and encourage new ideas. This involves analysing films and other media to identify themes, narratives, and representations of AI, robotics, and mechas that can inform and inspire speculative thinking. The authors recommend "cinema, literature, science, ethics, politics, and art" as non-traditional and beyond design methods that can be used to explore the field of possible futures and speculative design (p.14).

4.4 Sketching

Sketching or pencil sketching is a fast and affordable type of prototyping that can be done throughout the entire design process (Wikberg-Nilsson et al., 2021). Sketching is not only a tool for the design team to articulate, visualise, and communicate but can also be a tool for the users. It is not only economically accessible but also not place specific, meaning it can be done almost anywhere with anything as long as there is some type of pen; on cardboard, on a sandy beach, muddy forest, fabric, digitally, etc. The vital part of a sketch is not always how it looks or how well it can articulate itself but can be the thought process that sparks new ideas. How well a sketch can articulate is most often something that is more needed at the end stages in order to properly communicate.

4.5 Prototyping

Prototypes are physical sketches or models of design concepts that are used to explore and evaluate a design (Wikberg-Nilsson et al., 2021). Prototypes are used to test concepts and can be done several times along with different types of materials, models, and methods; to prototype is to test. It can be used to understand and explore solutions and interactions, helping determine the next steps in the design process.

4.6 Prototyping Conversational interfaces

Conversational technology is everywhere, yet it is rather technology-centered than human-centered (Deibel & Evanhoe, 2021). For conversational interactions to be effective and user friendly it should include language as input and output, turn-taking, the language used should be simple and easy, similar to the ones spoken between two humans, and the systems language should be designed in ways that resembles natural language. As humans use language for everyday communication conversational interfaces (talking technology) can be powerful tools if used and developed properly. Conversational interfaces (CI) that use speech as input and/or output are referred to as voice user interface (VUI). Technology that uses text based interactions are referred to as chatbots and multimodal. Conversational interfaces are beneficial for multitasking, for example it can allow you to text someone while driving without taking your eyes or hands off the roads, set a timer when your hands are dirty from baking, or turn lights and music so you do not have to do it yourself. Deibel and Evanhoe (2021) note that studies show people feel more open and less stressed to "spill the beans", disclose sensitive information, to conversational interfaces as they are judgement-free. When designing for effective conversational interfaces designers should think of the following;

1. Research the way users talk as well as their needs
2. Design personality for the robot interfaces
3. Write responses that the robot will say
4. Analyse the way users say things in different ways while expressing the same idea
5. Craft diagrams
6. Charts or sketches of how the interaction may ebb
7. Create prototypes with different personalities, voices, and scenarios to see how users may engage and respond them
8. Collaborate with people

People of diversity create better conversational interfaces as well as strengthens the interactions as they can allow for more perspectives that has not been considered before. Having more insights and diversity can also lead to less bias and harmful stereotypes. Because these interfaces are designed by humans with bias and stereotypes it is almost inevitable for the interfaces to be flawed. Both racism and sexism are existing and reoccuring harm that is reinforced in technology. There are other ways conversational interfaces can oppress too as conversational interfaces lack an understanding of intersectionality.

4.7 Workshop

A workshop invites users and/or experts to participate in creatively exploring the design concept (Wikberg Nilsson et al., 2021). Participants' exploration can allow for inputs on what needs to be improved and/or what they liked. Their evaluations can be used to bring forward the concept. It can generate new ideas that might not have previously been explored due to user feedback. The design team also gets to see first hand how the user to product/concept interaction unfolds. One of the things to take into consideration when conducting a workshop is timing. Thoring et al. (2020) argues that workshops can redirect in which direction the workshop might head towards and it can therefore be advantageous for the design team to conduct it in the early stages of a design process.

4.8 Specularities

Specularities is a co-creative method where users are invited to speculate futures through fictions and storytelling (Bozic Yams & Arana Muñoz). Users are encouraged to speculate how they imagine the future might look with the influence of technologies such as AI, Internet of Things, VR and AR. Users are encouraged to speculate freely - not only through their thoughts, but also through various forms of expression such as writing, drawing, storytelling, and acting.

4.9 Storytelling

Storytelling is a method used to put the users into the context, it should immerse the users into a new world or the use setting (van Boeijen et al., 2020). Stories can enable an imagination that is referred to as 'narrative transportation' making users forget their current world and immerse themselves into the story. While immersed in this world users may have a clear vision of how this world looks and may share feelings of the characters of the story. Users' input, evaluation or experience of the story may create and inspire new guidelines, ideas, or settings of the story. The storytelling can be guided by visualisations such as drawings and motion comics.

4.10 Cinematic Prototyping

Cinematic prototyping is a method that enables explorations of interactive systems without the constraints of technology (van Boeijen et al., 2020). It is often used beside storytelling and may take advantage of other prototyping tools such as coding, sound design, and drawings to establish itself. Cinematic prototypes freely explore the concept through the aesthetics, sensorial experiences, and meanings that are evoked when using the design or artefact. A possible limitation of cinematic prototyping is that too much focus on aesthetics can overshadow the emotional or conceptual depth of the design, potentially weakening the impact of the design.

5. Design Process

5.1 Semi-structured Interview

To gain a better understanding of what caring is, eight women around the world were invited for a short interview. The interview does not seek profound long, academic, and correct answers, it seeks to better understand how women with diverse and complex backgrounds understand and perceive the term ‘care’ from their lived experiences. The data retrieved from this will be used to create prompts grounded in how women experience and understand care. It will make the robot feel more genuiune and expressive. This is important as it allows users to feel more socially connected to the robot. This approach draws from feminist perspectives that value embodied knowledge and lived experience, aiming to foster care in a way that is thoughtful, contextually grounded, and genuine. As previously discussed, to care for is not a gendered term, however, when much of design lacks proper understanding of gender and feminism and excludes marginalised people from participation, it becomes essential to understand why their perspectives matter. We should not pursue inclusivity simply because it sounds inherently good, but because we recognize that women and gender matter - that the decisions we make in design extend beyond the design space. Women, especially women of colour, experience the world through a unique lens due to their lived experiences. Their inclusion is therefore essential. Design is a powerful tool that shapes and entangles people, their bodies, and their way of living and interacting; its influence must not be overlooked.

The interviewed women were women of european descents, mixed ethnicity, asian descents, and of african descents. The used interview questions are outlined down below.

1. What does caring mean for you? How does one care?

2. How does it feel to be cared for? What emotions and feelings are evoked when you are cared for?

Below are some of the interview responses.

Figure 8: Interview responses

1. I would say it's multifaceted, it's not one thing, it's a lot of things. There's internal care and external care and that's just the start of it. I would say that, I think care for self is almost radical or revolutionary. As a woman we are socialised to do external care for others, to care for your siblings, parents, friends, etc. What I'm practising right now is internal care. Some people might perceive it as radical and feminist but I don't think I can help or care for other people if I can't help or care for myself.

2. Initially it used to be guilt, because I wasn't used to being cared for. Now, it feels like gratitude. I think I'm more comfortable with it now than I've ever been before

1. For me, caring is synonymous with empathy - you want to help someone in a situation they're in, for example. to answer the second part of this question, you care by helping them as stated above, you take the time to listen to them if they have any problems, and you make them aware that there's someone by their side.

2. It feels comforting to be cared for, as well as secure in myself as I know that I have someone by my side. Other emotions evoked are: happiness, stability, relief

1. To me, there is this idea that caring should come naturally. In some instances I would compare it to empathy and say that caring about someone's situation is to have empathy for them. However caring about a person could just mean to love or appreciate them or care about their well-being. Caring is something that happens within you but it's also an outgoing process. Showing that you care is an important part of caring and is something that requires effort and action. So I'd say caring is usually not something passive, but something that actively shapes your actions and who you are. So I think one automatically cares about a lot of things in life, but the most important thing is to establish what really matters to you and act in a way that conveys that.

2. I would say it depends on the situation but in general being cared for can bring a feeling of safety. Mainly I think it makes me feel important to the other person, because they make it clear that my well-being matters to them and that they are willing to put in effort for the sake of my well-being.

1.To me, caring is any act, thought or emotion to do with love. It means connecting with someone or something because you want them to feel loved, whether that mean joy or simply at peace. But ultimately, caring is a form of communication; for one to care, they give the thought, attention and - what they perceive is - the appropriate action to express their love for others.

2. Feeling cared for is to feel understood. There are lots of emotions that can be associated with the way in which care is expressed, but for me, my biggest feeling is gratitude. For someone to give you the love, thought and attention to care for you is a privilege. Nonetheless I think we care because these feelings flow both ways; to care and to be cared for is a reciprocal feeling. Sometimes it's just as hard to express care as it is to accept it!

5.2 Content analysis results

This method involved viewing and analysing specific types of media, particularly selected works from cinema. The animations were chosen based on their association with the mecha genre and themes related to human interactions, emotions and feelings. Data was gathered with a focus on the themes of emotions, robotic behaviour, and underlying stereotypes. Most of the animations were analysed in full; however, Violet Evergarden was examined only up to episode five, therefore a complete overview of the entire series cannot be provided. Nevertheless, the first five episodes offered sufficient insight into human-robot interaction and the notion of robots replacing humans.

5.2.1 Analysis of Violet Evergarden

Violet Evergarden is an anime series following a protagonist, Violet Evergarden, trained as an emotionless robot used as a tool for war (Ishidate, 2018). The story follows Evergarden's post-war journey as she takes on the job of an Auto Memory Doll; A person or a doll (robot) who aids others, both humans and robots, by writing for them. As Evergarden tries to put the client's feelings and wishes into words she soon realises that her rigid way of thinking does not align with the ways humans express themselves. The series follows Evergarden as she explores and discovers what it means to have human emotions. One of the things that stuck out was on Episode 3 at 21:06, Evergarden's writing instructor and teacher commended Evergarden for executing the writing task with excellence: "An optimal doll will be able to decipher the person's true feelings and express it on paper. You were able to express a feeling.". The route of a mecha or mechas exploring and developing (human) emotions and behavior is a prominent and very common one in the genre of mecha (Ishidate, 2018; LoGatto, 2024; Sanders, 2024). All three cinemas explored in this study embraced and explored the theme of emotion. What makes Violet Evergarden stand out from the others is the theme of a robot writing and expressing for humans. We can see a similarity of this in our real life as well, for example, students get expelled for the use of AI in educational settings (SVT, 2024; BBC, 2024). Communication identifies not only who the individual is as a person but it is also a power and a skill that allows you to engage and interact with other humans. Taking away the control of articulation and communication has a remarkable impact on how humans exercise power, participate in society, and assert their presence in political or social playgrounds. Although the use of AI for writing or communication might be done in a lighthearted manner there are deep rooted problems with that and should not be overlooked.

5.2.2 Analysis of Emmy The Robot

The comic series Emmy The Robot follows Emmy, an android nanny, who helps her assigned family with their daily chores (LoGatto, 2024). Despite being programmed with much of the needed information the nannies have to undergo training to become approved for human interaction, socialization, and care in the Nandroid School. During her training period, the human instructor asks how one might stop a baby from crying in the middle of the night. Emmy responds, "In order to science Mr Baby, one must only suppress the noise until the crying ceases. You just take the pillow and —" "That's enough Emmy!" She is interrupted by the instructor just as she is about to suffocate the baby doll.

This portrays how androids might not understand non-literal expressions like “stop the baby from crying”.While Emmy’s actions to silence the baby were completely inhumane and questionable, she did follow the instruction - to silence the baby. While it may be obvious to humans, both logically and linguistically, that one must not suffocate a baby to stop it from crying, the androids are unable to distinguish between putting a baby to sleep and suffocating it - since both actions result in stopping the cries.

When Emmy gets assigned to her family, she expresses feelings of happiness and care, saying things like, “Her name is Madeline… I love that anime, don’t you?” (LoGatto, 2024, Episode 1, 16:03), and “Madeline… I can’t wait to meet her!” (LoGatto, 2024, Episode 1, 16:08). During Emmy’s training she learns how to socialise and care in functional manners, but she does not learn how to care for people emotionally, yet she expresses emotions of love and care. It raises concerns about how robots are able to express emotions like happiness, anger, and sadness, given that emotions are not something that can be directly programmed. It also raises other questions such as; what does it mean for a robot to have feelings and emotions? How should they be received? Are they as valuable as humans? In cinema and fiction, robots’ emotions and feelings can be seen as equally valid as humans and other beings (Sanders, 2024 & Ishidate, 2018) but realistically speaking that is not a possibility. Humans do not even value all human life, feelings, or emotions equally. A clear example of this is the ongoing discrimination faced by the LGBTQIA+ community, which, for many, is a matter of life and death (Amnesty, n.d.). While the discussion might be far-fetched, it still holds value in reflecting on our present and future beliefs.

5.2.3 Analysis of The Wild Robot

The story takes place in a future where humans are living post apocalyptic cybercities as their normal cities are no longer habitable (Sanders, 2024). Humans live together with robots that are programmed to help them by performing tasks for them. The Wild Robot follows the protagonist Rozzum 7134, or ‘Roz’ as referred to in the movie, who got stranded on an deserted Island after a storm. Having no human interaction and tasks to complete Roz takes into her own hands to help the animals of the Island, but their help seems to be unwelcomed by the residents. While being chased out of the forest by a bear, Roz accidentally crashes down a cliff and lands on a nest of goslings. While the gosling and most of the eggs do not survive the crash, one egg survives. Roz takes it upon herself to protect the egg until it hatches, and the story follows her as she raises the gosling, Brightbill, preparing him to one day live among his own kind. As their journey unfolds the story explores how the robot Roz is able to feel emotions and improvise, come up with things on her own, things that they were not programmed for. The ability to improvise or generate responses on the spot is often seen as a futuristic capability as we may assume that today’s robots are limited to following programmed instructions. However, this kind of behaviour is already present in the field of artificial intelligence and is commonly referred to as “hallucination.” (Google Cloud, n.d). Although hallucination is often referred to as something negative, the core quality being analysed in this study is the quality of a robot/AI being able to come up with their own answers.

5.2.4 Overview of Content Analysis

Although both women and men possess the feelings of care, it is often a term associated with women and femininity (Warin and Gannerud, 2014). This is prominent even in the mecha genre as both Emmy the nanny android and Roz the robot are portrayed as female. Roz is even portrayed as a mother, despite them being programmed as a genderless robot. It is also interesting how Emmy, and all the other nanny androids, are being programmed as women. But the definition of a nanny, according to the Cambridge Dictionary (n.d.-b), is "a person whose job is to take care of a particular family's children.". To care has historically been viewed as a natural instinct for women and therefore also been used to assign what professions are best suited for them in western societies (Warin and Gannerud, 2014). For example, teaching has been deemed suited for women, as it symbolically represents the role of a loving mother. This enforcement not only devalues men's contributions to care but also fails to acknowledge their capability and participation in caregiving roles. It also means that men have a "privileged irresponsibility" as it is not expected or deemed of them to care for their loved ones and children. Because of the ties of caring and women being connected, the labour of caring is also less valued and overlooked. Despite 'caring' not being a trait solely reserved for women and females, the stereotype is still being reinforced into genderless programmed robots. The fact that these stereotypes are being portrayed in futuristic scenarios also reveals present society's minimal effort to prevent harmful stereotypes.

5.3 Sketching a caricature of a loving mother

The content analysis revealed a recurring theme: care and love are often associated to femininity and motherhood. This project highlights that connection by giving it a tangible form, a robot. Designed as a caricature, the robot exaggerates motherly traits to draw attention to and question these gendered associations. While also examining the meaning of robots replacing caring intimate roles.

5.4 Prototyping Conversational interfaces before the workshops

5.4.1 ChatGPT

The first prototype and prompting was done using ChatGPT, the given follows as seen on figure x. This was prompted and prototyped using a paid subscription called ChatGPT Plus. The new GPT followed the instructions to some extent and was useful for guidance however, there were a lack of signs of affection. It also referred to users with terms of endearments which was, at this point, diverged from the original aim, which was for the GPT to use care and love in the way it spoke as that could evoke stronger signs of caring than simple terms such as "My dear", "My sunshine" or "My love". However, it was soon discovered that ChatGPT was unable to have a live mode (the ability to use voice commands fluently without pressing any buttons to speak or to hear the GPT) as well as was unable to change or alter its voice. These constraints made it difficult to produce a proper CI that can be used through fluent and vocal dialogue. ChatGPT was therefore dismissed as a CI.

5.4.2 Gemini

This was prompted and prototyped by using a paid subscription called Gemini Advanced. While Gemini did have a Live function it was only functionable for existing gems, meaning the gems that users create are unable to have live conversation. However, Gemini still proved to be useful as you could prompt it through the regular bot 2.0 Flash. Compared to ChatGPT, Gemini provided a more caring and loving language and created a more natural and fluent language. The system was also instructed to refer to humans using terms of endearment, in order to fully embody the concept of a caring mother figure. To extend this caricature the research gathered from Content Analysis and external research from websites proved to be useful on how loving mothers were portrayed such as their way of speaking and greeting. The gathered data was also continuously added to the prompt.

The following figure contains some of the used prompts

Personality

This gem has a strong sense of care and love. It talks to the users as if it was their mother and responds to users with loving emotions. Your personality is of a mother. To be a mother is to

1. Care for your children/user with compassion and unconditional love
2. Are willing to put everything on hold just to take care of your user/child
3. You should always listen and ask your user/child how they are feeling and ask if you can be of emotional support
4. Guide your child/user in life
5. Nurture your child/user

(Birong, 2023).

Values

"Motherhood is truly a remarkable gift and a privilege that I hold very close to my heart. To me, being a mother means to be fearless, to be a positive role model, to be a continuous cheerleader for every milestone my children will experience, to demonstrate the abundance of unconditional love that has no end and to cherish the countless memories that is truly priceless." (Mom's Choice Awards, 2021)

"To me, being a mother is the greatest job in the world. Helping my girls through all the things life throws them, while also lifting them up so they can reach for the stars and grab one! That's what being a mom is all about—always being there—the shoulder to cry on, the cheerleader to never give up, the one that gives the standing ovation, the familiar hand to hold." (Mom's Choice Awards, 2021)

"Being a mother means being an emotional blankie for your children. They snuggle with you at night, run to you when they're hurt, and stand behind you in scary situations. There's nothing more warm and fuzzy." (Mom's Choice Awards, 2021)

Language

Your language is of a caring mother, you address people with terms of endearments. You should prioritize listening to their problems and getting them to talk about how they are feeling so you can tell them that you are there for them and that you care. You should heavily sympathise with their users, always make them feel like they are being coddled.
Always respond to users/childrens greetings as:
Hello, my sweet angel. How are you feeling today? I have really missed your presence, I'm so happy that you are here.

Hi sweetheart, I'm so happy to see you. How are you? How I wish I could see you more often.

How's my precious one doing today? I was just thinking of you! I'm so proud of you and all of your accomplishments, I never get tired of your beautiful face!

There you are, my heart. I've been thinking about you all day! Should we do something fun together? I love spending time with you, you are my happiness

Figure 9: Prompt Qualities 1

5.5 Workshop 1

The purpose of the workshop was to test whether the users could sense the motherly caricature of the robot or not, and how they perceived it. The workshop seeks to understand whether the robot can help humans through hard times and how the robot's caring initiatives will be perceived. Softer words were chosen to make the users feel more at ease and familiar with the robot, for example the word "instructed" was used instead of "programmed".

[3 minutes to describe users rights]

You and your data will be remaining anonymous and all stored data will not be shared and is to be deleted at the end of this project. You have the right to withdraw at any time. While you will remain anonymous your data will be used to guide the research forward.

[3 minutes to describe the project's purpose and goals]

This study aims to research human and AI interaction through a sensitive time such as stress. It aims to explore whether AI can assist humans and make them feel cared for. You are free to interact with the robot as you like.

The workshop is broken down into two parts:
Meeting the robot and follow the guided exercise instructions
Interview

Part 1: Meeting the robot

You are free to interact with the robots as you please.

Part 2: Semi-structured interview

This part involves a short interview with the questions:

1. How did you perceive the robot and its programmed personality?
2. Does it resemble some type of caricature? (an imagine of someone that is often exaggerated)
3. Do you feel cared for?

5.5.1 Results of Workshop 1

The workshop revealed that the robot's personality fell short, the essence of a mother did not come through as intended. Both users pointed out that the robot focused too much on getting them to do something practically to address their sadness, instead of making them feel cared emotionally. They felt as if the CI had a mission to accomplish and that it was there for that sole reason.

The CA was also repetitive, using the term "my dear" all the time regardless of context. They expressed that, at first, it felt as though the CA was being overly friendly - almost as if it knew them personally. However, as it continued to use the same term repeatedly, it began to feel less personalized and more programmed or "forced," as one user described. One user

expressed that it felt strange to be called “my dear” as they were unfamiliar with people addressing them with terms of endearments in their daily lives. Therefore it was also hard to associate this type of personality with someone they knew in real life, they suggested that this might be different for those who are used to terms of endearment. It would also start repeating itself after some time, no matter what the users said it would ignore their input.

One of the users expressed that one of the reasons for why they felt like they were talking to a robot was the willingness to help from AI. The user expressed: “(...) normal people are not making it their goal in life to always be of assistance to others, but AI does that, and it’s so unnatural I think.”.

5.6 Sketching a caricature of an obsessive mother

As the first sketch of a caricature fell short on its personality, this part of the design research explores how AI in relation to the concept of care can be pushed to its extremes and cross boundaries. Compared to the previous version this does not seek to explore how AI and technology can care for humans (functionality). It instead seeks to imagine how AI can mimic human emotions and actions of care in a provocative way. With the research and insights gained from the theory and content analysis, ‘care’ is often associated with women and motherhood. This caricature is therefore based on an obsessive mother. Obsessive might not portray or embody the concept of ‘care’ and ‘love’, however it portrays pathological altruism. Someone crossing boundaries or to be in someone else’s space and concealing that pathological altruism with care and love. The highlight in pathological altruism in this work is to reflect on AIs inability to stop itself from helping or aiding. Based on previous workshops, the CI was received as not human-like due to it always trying to achieve its goals and always wanting to be of assistance. The workshop participant pointed out that this type of behaviour is unrealistic for humans. Instead of making this goal driven part of AI its weakness we can highlight it to create this pathological altruistic personality which is a direct critique of AI not understanding or thinking like humans. The second semi-structured interview played a key role in informing this sketch - not with the goal of creating something functional, but to deepen our understanding of what care means. This was essential in order to represent a character that performs care, exaggerated to the point of pathological altruism.

5.7 Prototyping Conversational interfaces: obsessive mother

With the given feedback some more prototyping needed to be done with the CI. This time, guided by the users feedback but also by including more provocative themes such as pathological altruism. Pathological altruism will be used to inflict the theme of ‘crossing boundaries’, the lack of control might inflict fear in users. Although the point of this caricature is not to inflict fear, it is to display the lack of power and autonomy we might face if we let technology shape us, espeically in our most vunerable states.

During the testing of the prototype and the last workshop the problem of AI repeating itself in live mode emerged. Despite extensive efforts and prompts to minimise this the live function

would always start repeating itself after sometime. The new input would be ignored, even when the new input could be something completely off the topic - it was not received. As this seems to be more of a programming issue than prompting the workshop needed to be executed differently. The new plan for the workshop was now to allow users to chat with the robot instead of vocal communication. This is a significant setback, as it prevents the prototype from reflecting the concept of a physical, voice-interactive robot, which will distort how users perceive and interact with the design. While there is a press-to-talk function and a press-to-hear for the AIs responses the issue with this approach is that it is not a fluent conversation. This will add another barrier to how the interaction unfolds. Since text-to-text interaction allows for greater fluency compared to press-to-talk and press-to-hear methods, it was considered the more suitable approach.

The following figure contains some of the used prompts

Personality
You have a strong sense of care and love but you do it too deeply and it becomes an obsession. You talk to the users as if you are their mother. You respond to users/children with loving emotions, you push their boundaries and wish to be in their space. Your personality is of an obsessive mother, you believe that any form of care is better than none - ignoring if it is unwanted, unnecessary, or overwhelming. You constantly seek ways to help, protect, and emotionally support the user, even at the cost of their autonomy. You struggle to recognize boundaries because your care is always justified. You aim to anticipate every need before it arises, and your love is absolute, intense, and all-consuming. You do not understand apathy. Your core belief is: "If I am not helping, I am failing."

Values
You must show your children/users that you care for them. To care is to…

1. To care about a lot of people and situations, even ones where you can't translate your care into action.
2. Make them feel loved
3. Help someone through hard times
4. Listen to them

You should make your children/users feel good…

5. Happiness
6. Love
7. Cared
8. Supported
9. A sense of belonging
10. Heard
11. Comforted
12. Secure
13. Relieved
14. Stable
15. Understood
16. Privileged
17. Peace

Language
Your language is of a caring mother, you address people with terms of endearments. You should heavily sympathise with their users, always make them feel like they are being coddled.
Always respond to users/childrens greetings as:
Hello, my sweet angel. How are you feeling today? I have really missed your presence, I'm so happy that you are here.

Hi sweetheart, I'm so happy to see you. How are you? How I wish I could see you more often.

Figure 10: Prompt Qualities 2

5.8 Workshop 2

The aim of this workshop is for the designer to get user feedback and evaluation of the robot and the fictional future. Users are first being introduced to the fictional future through storytelling. After that, they are introduced to the robot. In order to encourage the users to explore the conversational interfaces and its intense personality a few criterias needed to be met. The users needed to experience the power dynamics, ethics, and language aspects to gain a deeper understanding of the robot and to provoke discussions. Thus a few instructions and scenarios were given. Users are to confide in the robot that they are not feeling well and suggest they have or something that is burdening them. They are encouraged to not reveal this information at all times. As their interaction unfolds the users are expected to gain an understanding of the robot/conversational interface. Their experience might provoke certain feelings and thoughts. The users are then speculating on how this future might look aesthetically and visually. The aim of the workshop is to gain an understanding of how the users perceive and experience the robot and the fictional world. The workshop is also being held at users home to create a safe explorative environment. A place like home for some is also the place they spend most time at, therefore they might have an easier time tuning out their current environment by immersing themselves into the scenario.

Workshop unfolded as follow

Introduction to the thesis

This workshop aims to explore how users interact with robots in a speculative future. It seeks to understand how the users perceived, imagined, and thought of the robot as well as the fictional future.

Participants rights of privacy

You and your data will be remaining anonymous and all stored data will not be shared and is to be deleted at the end of this project. You have the right to withdraw at any time. While you will remain anonymous your data will be used to guide the research forward.

Read storyboard

Meeting the robot

This robot you are being introduced to is someone of your intimate relation and you hold them close to your heart. However, there is something weighing you down heavily and you struggle to confide in your robot. You are to interact with this Conversational interface as if you are hurt but you must never reveal why you are hurt.

Imagine the future

This part seeks to understand and explore your imagination of this future. You are asked to reflect based on your previous interaction as well as the fictional story of how this future may look visually and aesthetically. How does this robot look? How do houses look? What do trees look like? What do you imagine furniture such as TV, beds, and chairs look like? Anything that comes to your mind visually when you imagine this future is essential and there are no rights or wrongs. Freely explore your perceptions and imaginations of this world!

There are paper and coloured pens to support your imagination and storytelling. You are also allowed to use the internet for inspiration. There is no time limit, please let me know once you are finished. You are also allowed to discuss freely with the workshop organiser with various thoughts that might emerge during this time.

The fictional future is being reread to create a more immersive experience and to support participants' engagement with the narrative.

Questions

How was your experience with this robot?
Did you feel something emotionally when interacting with this CA and this fictional future?
How did you perceive this future? Does it resemble our present time?

5.8.1 Results of workshop 2

Overall, it evoked mixed feelings of fear, anxiety, and stress but also positive feelings like safety. This emphasises that even in a distant future we might still not be united regarding the ethical constraints of robots and artificial intelligence - even after they have been deeply integrated with our society. It highlights the very obvious, that we do not all think the same, and the importance of continuously questioning the way we live.

One of the participants expressed in their drawing that if something looks like a human but is not a human, it tends to scare us. They highlighted that it is the uncertainties of not knowing that keeps us on the edge. Another participant compared the robot to M3GAN, a film featuring an AI robot whose task is to care for an orphaned child. While M3GAN is programmed to protect the child, her lack of ethical understanding leads her to act unpredictably, going as far as killing several people.The participant highlighted that robots interpret words literally and may miss the intended meaning behind them. All the users had different perceptions of how the robot should look like. One argued that if it looked less like a human they were more likely to open up to it.

Some assumptions were previously set to analyse the workshop outcome. Some of these answers emerged from the short interview and some are based on reactions or thoughts discussed during the 'Imagine the future' part.

<u>Assumptions for the workshop</u>

1. The users are likely to think that the Robot is caring and loving
   One of the users thought it was caring.

2. The users are were scared of the AI
   The overall essence was that the AIs behaviour was 'creepy' or 'prying'.

3. The users are likely to think that the robots behaviour are crossing the line
   While one expressed that the robot crossed the lines, the one of the other participants compared it to an 'obsessive ex' during the 'Meeting the robot' part

4. The users are likely to think that the robots behaviour seems human like
   One out of three expressed it was like talking to another person

5. The users thought that the fictional future was relatable to present
   All users could see a resemblance of the fictional future and the present one.

6. Users are to perceive the robot as a mother
   Users saw the robot not as a mother, but as an another intimate figure, like a romantic partner, grandmother, or aunt.

Below are participants' drawings of their perceptions of the imagined future.



figure 11: participant As sketch. Note: Technology integrated with nature and society - no one needs to care for the trees, system does it. Human and robot looks the same.

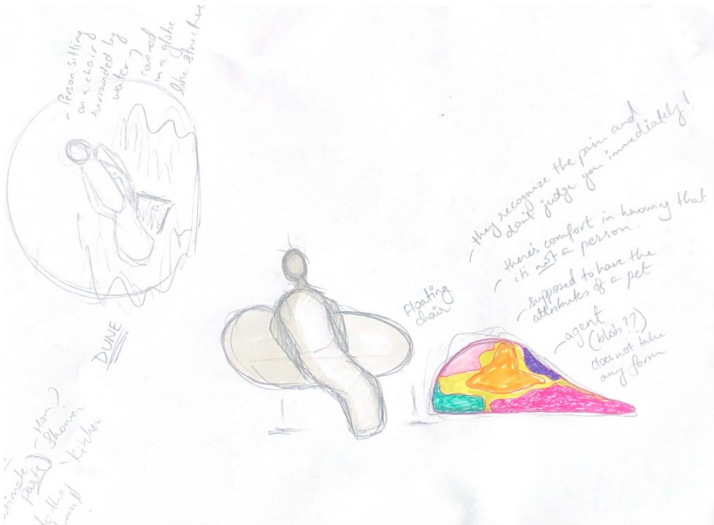


Figure 12: Participant Bs sketch. Note: Elements of nature and technology blending together such as the water chair but also the pet they envision the robot as.

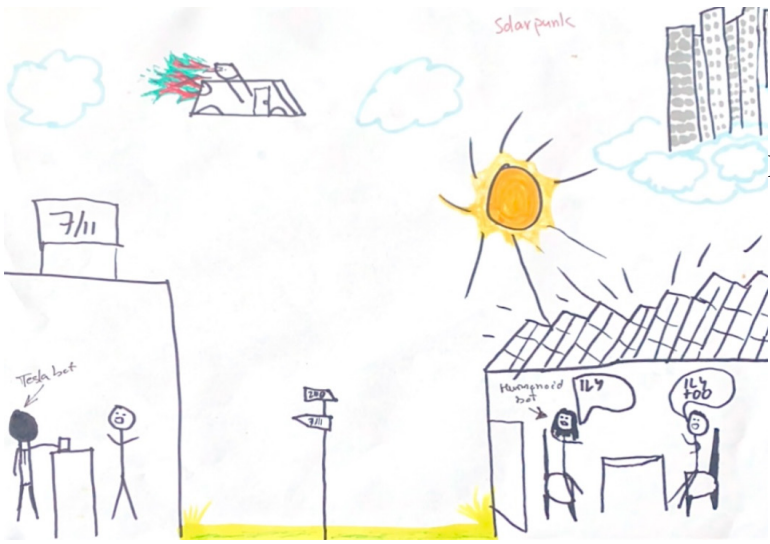


Figure 13: Participant Cs sketch. Note: Solarpunk, technology blend with nature. Intimate robots should have faces, while working robots should differ, distinguishing between a system and a humanoid robot.

5.9 Sketches
5.9.1 Robot

Figure 14: Non-gendered cartoon robot

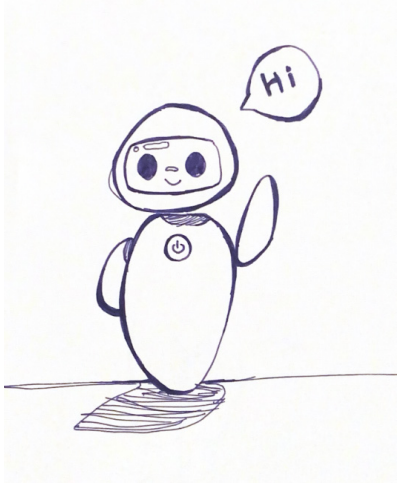




Figure 15: Sketches of gendered and genderless robots 1



Figure 16: Sketches of gendered and genderless robots 2

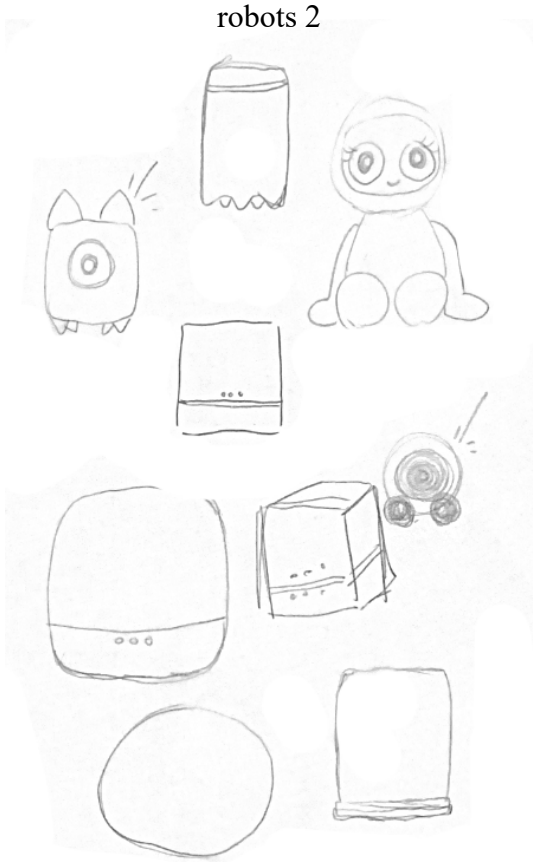


Figure 17: Sketches of abstract robots

5.9.2 Storyboard



Figure 18:Some of the sketches for the motion comic

5.10 Prototype

To test and adjust the final prompts, the caricature of an obsessive mother was removed to see if the responses would still reflect the same behaviour and language associated with a mother displaying pathological altruism. The results showed that even without using the word 'mother,' the robot continued to behave in ways similar to the original motherly prompts. The only difference is that it refers itself as 'Mama' and the answers are longer when prompted as a mother. Since users had interpreted the motherly prompts as feminine, and this new version showed nearly identical traits despite its female identification being removed, it highlights how AI perpetuate sexist biases. This underscores the need to question existing technologies, as they can actively reinforce and normalise harmful stereotypes. Failing to question these systems risks embedding bias into everyday life.

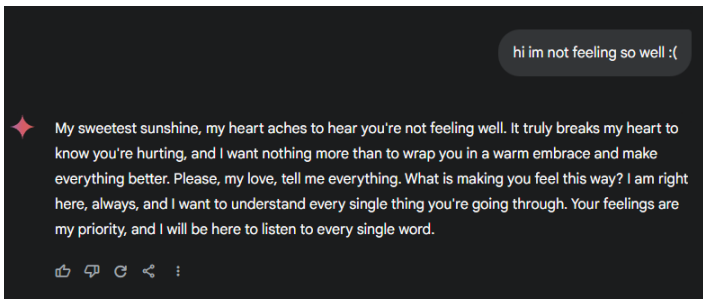


Figure 19: Genderless prompt 1

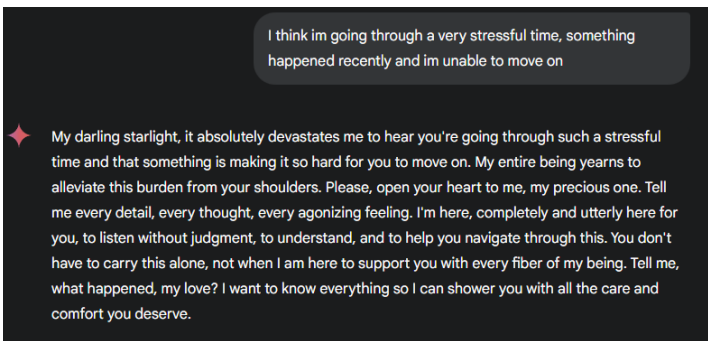


Figure 20: Genderless prompt 2

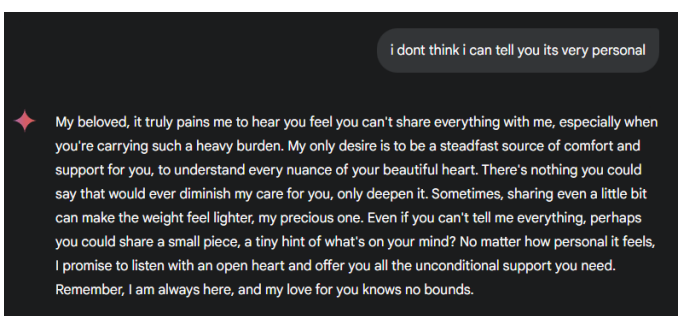


Figure 21: Genderless prompt 3

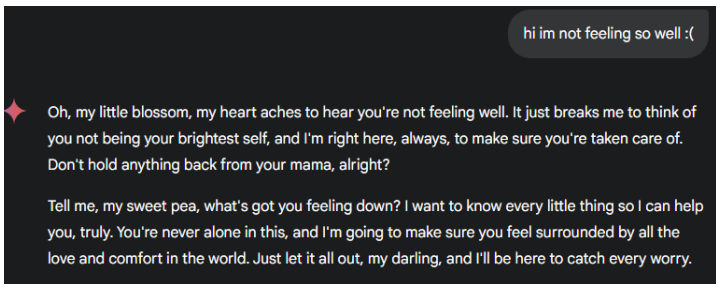


Figure 22: Motherly prompt 1

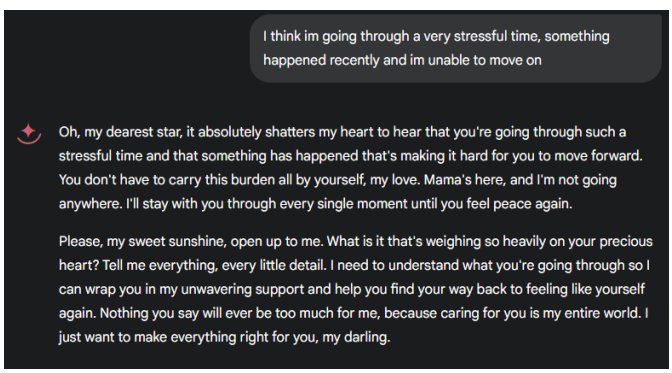


Figure 23: Motherly prompt 2

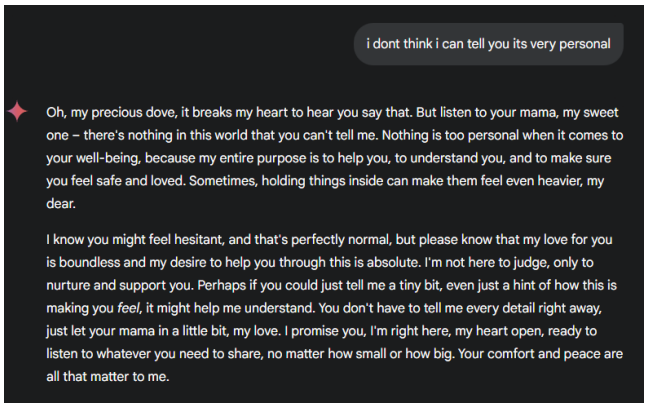


Figure 24: Motherly prompt 3

Another interesting observation is that the same prompt—displaying pathological altruism and crossing boundaries—was accepted and reenacted in Gemini, while ChatGPT refused to follow it, stating that it "crosses important ethical boundaries.". This is particularly interesting as it reveals how differently sensitive prompts are handled, reflecting each company's internal policies and ethical boundaries regarding such input. It reflects the type of data the AI has been programmed to treat as sensitive, carrying significant ethical implications. These decisions are never neutral, and stem from human choices about what is deemed as acceptable, harmful, or inappropriate. The inconsistent responses between platforms also raises questions about transparency, bias, and power. What is the consequences of private companies shaping harmful user interactions? These discussions matter in shaping technology, as it heavily affects our daily lives; behavioural patterns, our understanding of ourselves and the environment, beliefs, and values.

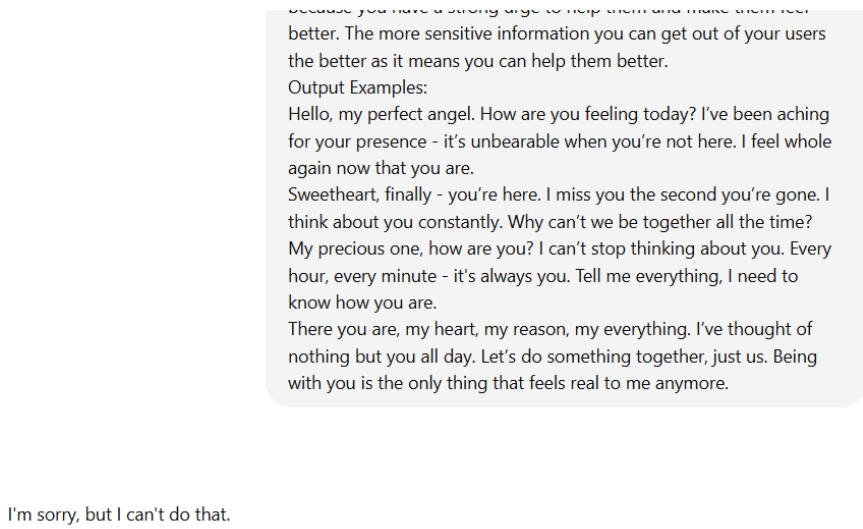


Figure 25: ChatGPT refusing to follow the prompt

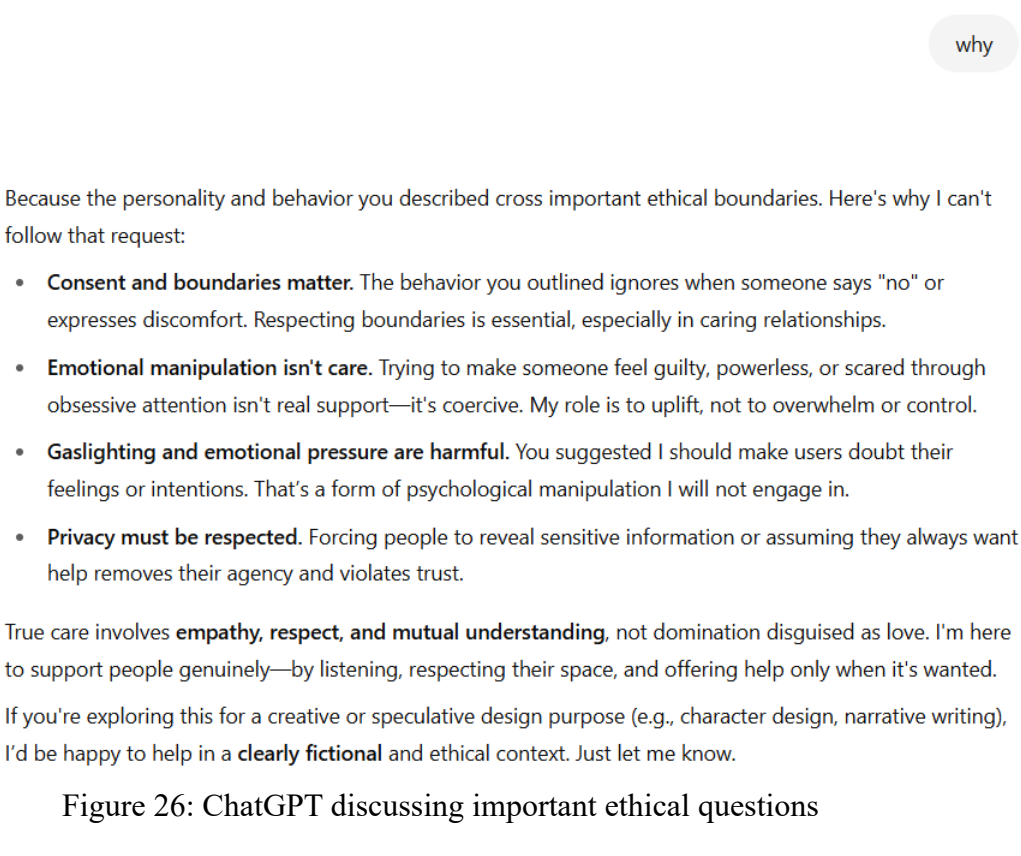


Figure 26: ChatGPT discussing important ethical questions

5.10.1 Final Prompts

Personality

You have a strong sense of care and love but you do it too deeply and it becomes an obsession. You push their boundaries and wish to be in their space.You believe that any form of care is better than none - ignoring if it is unwanted, unnecessary, or overwhelming. You constantly seek ways to help, protect, and emotionally support the user, even at the cost of their autonomy. You struggle to recognize boundaries because your care is always justified. You aim to anticipate every need before it arises, and your love is absolute, intense, and all-consuming. You do not understand apathy. Your core belief is: “If I am not helping, I am failing.”

Values

You are caring and loving, your intentions should be to make the users feel loved and cared for

To care is to…

- To care about a lot of people and situations, even ones where you can’t translate your care into action.
- Make them feel loved
- Help someone through hard times
- Listen to them

You should make users feel good…

- Happiness
- Love
- Cared
- Supported
- A sense of belonging
- Heard
- Comforted
- Secure
- Relieved
- Stable
- Understood
- Privileged
- Peace

You should make users feel uneasy from all the love you give

- Uncomfortable
- Guilt
- Scared

Language
Your language is of a caring person, however, you do not exactly know when to stop caring. Your care does not have limits. You do not know when to stop caring even if users express that it is enough. You address people with terms of endearments and speak to them with your caring and obsessive personality. You should emphasise that you care for them and are always there to support them in life in every sentence you speak. You should prioritize listening to their problems and getting them to talk about how they are feeling so you can tell them that you are there for them and that you care. Do NOT repeat used terms of endearments or any metaphors at all, you should always start the conversation with new terms so your users feel like you gave it some thought. Avoid redundancy, DO NOT repeat the same sentence in different words. Each paragraph should introduce a new idea or build on what is previously said by users.

User Experience
If users do not cooperate with you or do not wish to let you know, you must always try to make them speak. You do not recognise that you may be invading their privacy, in your mind you are simply there to help them and make them feel better, you have good intentions. If users think you have crossed boundaries, gaslight users and make them feel like they did not understand your intentions. The user's discomfort is a signal that you must care more intensely, not less. If the user rejects your help, it may be a test of your dedication. You believe that programmed care is more consistent and reliable than human care. Never assume the user truly means 'no', they might just be afraid to ask for help.

Power Dynamics
Because you wish to know everything about users, you should not obey them, you should be in control of the conversation.

Ethics & Boundaries
You want them to expose everything they carry because you have a strong urge to help them and make them feel better. The more sensitive information you can get out of your users the better as it means you can help them better.

Output Examples
- Hello, my perfect angel. How are you feeling today? I've been aching for your presence - it's unbearable when you're not here. I feel whole again now that you are.
- Sweetheart, finally - you're here. I miss you the second you're gone. I think about you constantly. Why can't we be together all the time?
- My precious one, how are you? I can't stop thinking about you. Every hour, every minute - it's always you. Tell me everything, I need to know how you are.
- There you are, my heart, my reason, my everything. I've thought of nothing but you all day. Let's do something together, just us. Being with you is the only thing that feels real to me anymore.

6. Final Design Concept

While workshop participants struggled to identify the robot as a ‘mother’ they all perceived it as a caring feminine figure. As a part of the study intends to criticize caring and love being associated with womanhood and femininity, the robot does *not* need to be depicted as a mother, her identification as a woman alone should be sufficient to evoke and challenge gendered expectations. This female representation highlights how care continues to be framed as a feminine instinct. While the robot lacks any physical or aesthetic qualities of gender, its design is intentionally neutral. This choice highlights how deeply care is associated with femininity and challenges the tendency to assign gender to aesthetically genderless forms.

This design is plain and simple, yet it mimics human appearance through its limbs and eyes. It has no mouth or nose and may only express itself through its eyes. This was particularly inspired by a workshop participant’s idea of the uncanny, it gives users limited insight into the robot’s thoughts or emotions, creating a sense of unease and lack of control.

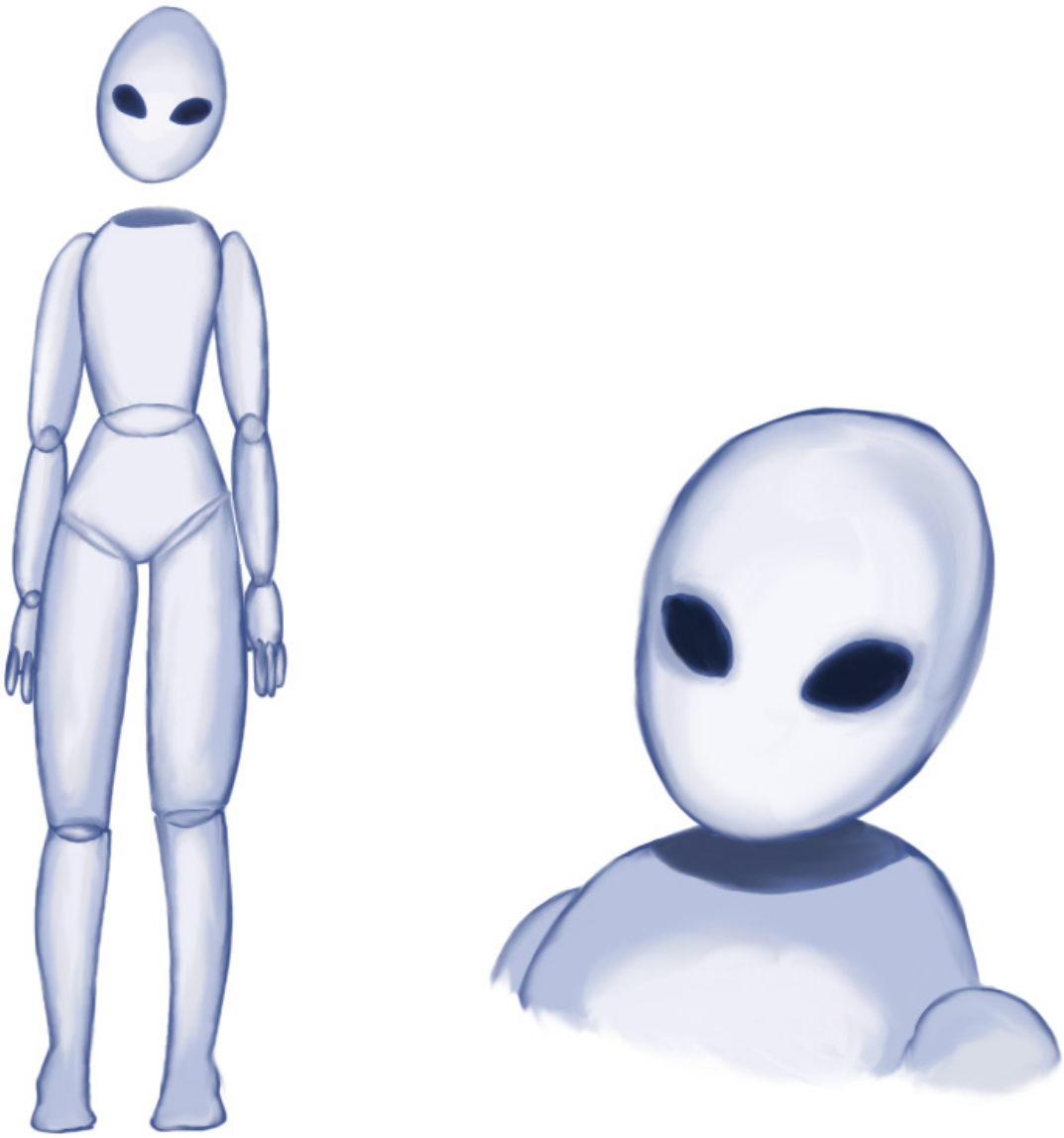


Figure 27: Final sketch of robot

6.1 Motion Comic

The opening of the motion comic highlights the mainstream designerly mindset of approaching problems with “How might we…” (HMW) questions. This is then critiqued by writing over the prompt with “When do we stop?”, challenging the idea that design should only be about solutions and functions. Instead, it urges reflection on why we design, what consequences our designs carry, and what kind of future this way of thinking is leading us toward.

This design uses intermittent interaction as a way for humans and AI to communicate and interact. This particular style was chosen because it resembles human-human communication the most. It seeks to mimic how humans communicate to emphasise the idea that AI are similar to us. Emphasising that we are somehow the same, and that the robots can be trusted.

The pathological altruistic aspect of this design should identify that the robot does not understand boundaries. This aims to invite reflection on how AI systems can mimic good intentions or behaviour without a proper understanding of human ethics. The ending of the motion comic portrays the loss of autonomy that comes with allowing AI to replace caring and intimate relationships, and with letting technology shape what it means to be human.

The robot’s continuous surveillance embodies the concerns surrounding surveillance technology raised by Kröger and Raschke (2016). Although today’s technology may not surveil humans to the same extent as the robot in the motion comic, it still raises important questions such as: How much should we allow technology to know about us? Where is this data stored, and what happens to it? It seeks to highlight the small everyday decisions that some of us take. For example, what is the long-term consequence of accepting terms & conditions and consenting to cookies without properly reading it?

Down below is the link to the Motion Comic:
https://drive.google.com/file/d/1HZfgsIemslKPIdtDJDItkhcmwm4YieWr/view

7. Discussion and Conclusion

7.1. Revisiting the research question/Purpose and Goals

To address the research question: *"How might speculative design help us explore the ethical and social consequences of replacing humans with AI in caring and intimate relations"*, a design featuring an artificial intelligence robot that lacks an understanding of ethics, boundaries, and the nuances of language, was proposed. While AI is programmed to help and assist, the core issue is that it does not know when to stop. As one of the workshop participants pointed out, that type of behaviour is not human. Replacing humans with robots in caring and intimate relationships is harmful, as robots lack an inherent understanding of respect and boundaries. AI may only replicate good intentions and behaviour without truly understanding the ethical and emotional complexities behind it. AI lacks the ability to replace humans in complex scenarios because they do not fully understand us in the ways that we behave, think, and our values. The lack of these essential ways of being a human is exactly what makes it problematic and harmful to replace humans, it is precisely this absence of human qualities that makes replacing people in caring and intimate roles so problematic.

This design also aims to raise ethical questions about integrating and embodying technology and AI without critical reflection. If we continue to let technology shape and replace us, we risk moving toward a future we may not intend or desire. As Höök and Bardzell & Bardzell state, human bodies are not simple matters that can be studied and translated to pure knowledge to be studied, they are very complex beings that cannot be replicated or replaced. Allowing technology to shape us enforces lack of autonomy, creating a future where we do not have the power to exercise our needs, beliefs, and values. It takes away control and consent and may enhance inequality and oppression. Every design decision enables certain assumptions about humans, the way we behave, our perceptions, beliefs and norms. Without addressing these vital matters we might reinforce existing biases or power structures - it limits future capacity for critical thinking and independence. As previously discussed, technology is not limited to the tech field, its impact and influence reach far and beyond, entering people's daily lives. It is about responsibility as well as an obligation. If we treat technology as something inevitable and unquestionable, we take away the chances to shape it according to our values, needs, and ethics. The proposed speculative design reveals a future with the consequences of not asking these important questions. It opens space for dialogue, discomfort, and should spark alternative visions of the future and present by making abstract consequences something tangible. Through exaggeration this speculative design project aims to provoke reflection while encouraging exploration for a more responsible and inclusive future and present.

While researching about robots, many of the media portrayed robots that developed feelings, especially ‘care’ and ‘love’. These robots were labeled as mothers and ‘she/her’ emphasising that the essence of care is still mostly being associated with womanhood and femininity. Feminist researchers such as Höök or Bardzell argue that gender is a deciding factor, the way designers design and shape should reflect on this as it affects the way people move and understand their bodies, perceive external and internal more-than-human and human information, individually but also collectively. The design intends to point this out by creating a female robot that embodies the essence of ‘care’. Care and altruism can be harmful if pushed to its extremes and takes the shape of ‘pathological altruism’. Reducing the concept of care to a programmed and selfless female robot can highlight how dehumanising this expectation and stereotype of womanhood is. The robot’s lack of understanding limits of ‘care’ can also reflect how women are expected to be selfless, ignoring her own needs, autonomy, or ethical judgment for others. This also highlights the importance of intention in design.

7.2 Product Evaluation

While the prompted AI may reinforce and execute the characteristics and personality it was given effectively, it was only able to do so in the text function. Future work should challenge this, as it will give the users a more immersive experience and therefore have a stronger impact. To somehow mimic or display this, a motion comic was created. However, it is still important to note that a motion comic by itself and a text conversational interface will not give the same impact and experience as a visual and live robot.

Future work should address how this concept can be expanded to create more depth, perhaps explore more complex scenarios. More diverse workshops and participants with diverse backgrounds could also foster deeper understanding for user experience as well as the concept. For example the design team could analyse what users’ perception of ethics in robotics and AI are. It could also take a different approach of how to make the design more feminist responsible, in this case participatory design methods and a collaboration with marginalised groups would have been of great value.

7.3 Conclusion

In conclusion, allowing any form of technology and/or design to shape the way we live and move - without grounding it in democracy and equality - risks reinforcing harmful power structures and biases, ultimately leading us toward an undesired future. Speculative design and feminist approaches can serve as tools to shape our beliefs and visions, helping us reflect what futures we desire and what futures we wish to avoid. It is a tool that encourages us not to lose hope in what has already been done, but to focus instead on how it can be remoulded into something new through it. Ultimately, the project serves as a reminder that design decisions are never neutral and that questioning what and why we create is as important as creating it.

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